BROADMIND AI • AUTONOMOUS ACADEMIC ENGINEERING

Master Architecture & Engineering Dossier

Complete Technical Blueprint: Methods, Cognitive Algorithms, All 17 Modules, API Cascades & Design System

PLATFORM VERSION	FRONTEND ARCHITECTURE	BACKEND DATABASE	CORE COGNITIVE AI
BroadMind AI 5.2 Enterprise	**Next.js 16 (Turbopack)**	**FastAPI + PostgreSQL (DB_BM)**	**NVIDIA NIM Nemotron 3.5**

1. Platform Executive Overview & Mission

BroadMind AI ("Read faster, write better, code smarter") is a next-generation autonomous academic platform built to serve learners across competitive Indian exams (GATE, JEE, UPSC, NEET), engineering, medicine, and higher education. The platform unites 17 interactive learning modules under **one shared adaptive engine**. Rather than running disconnected heuristics, every module streams standardized telemetry events into a centralized event store, and queries the engine for real-time pedagogical decisions, mastery estimation, and complexity routing.

> **The Dual-Engine Foundation**
> BroadMind operates across two tightly coupled tiers: (1) A high-performance **Next.js 16 React Web Client** running at 120 FPS with high-DPI canvas physics, and (2) An official **FastAPI / SQLAlchemy Backend (DB_BM)** operating on port 8001 with dual-dialect PostgreSQL (pgvector) and zero-config local SQLite support.

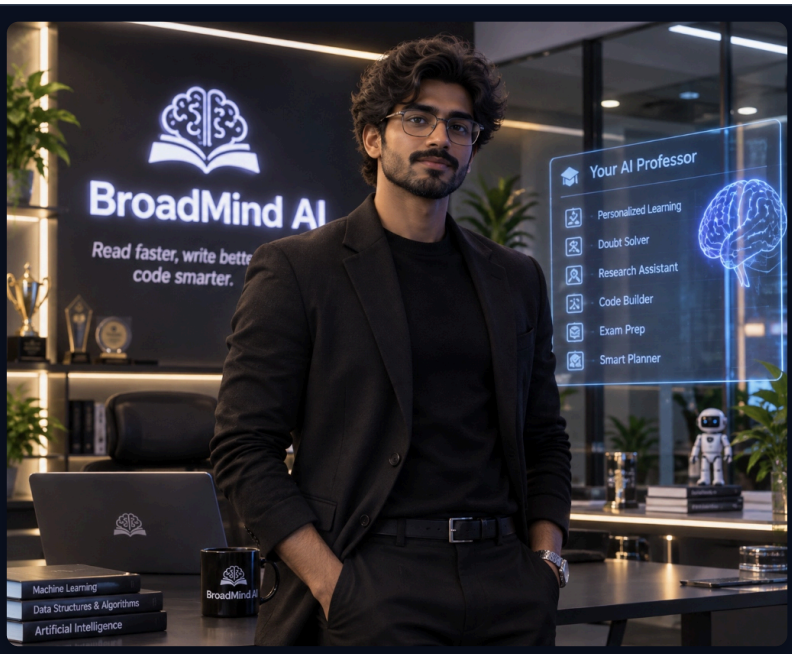


Figure 1.0: The Official BroadMind AI Universal Professor with 3D Floating Holographic HUD ("Your AI Professor") displaying the 6 Core Pillars in the BroadMind Classroom.

2. Design System & Complete Color Palette Specifications

The visual architecture of BroadMind AI follows a purposeful "Dark Academic Executive" aesthetic anchored by the official **White-Green** brand identity. High-contrast typography, mathematical chalkboard tones, and neon-cyan HUD elements create focus without cognitive strain.

COLOR NAME	HEX CODE	TAILWIND & CSS TOKENS	SEMANTIC ROLE & PSYCHOLOGICAL PURPOSE
Obsidian Slate	`#020617`	`bg-slate-950` / `--bg`	Primary canvas background. Minimizes eye fatigue during prolonged night study sessions.
Deep Navy Card	`#0f172a`	`bg-slate-900` / `--card`	Surface containers, sidebars, modals with `backdrop-blur-md` and 1px border.
BroadMind Emerald	`#10b981`	`bg-emerald-500` / `text-emerald-400`	Primary brand identity accent. Represents cognitive growth, progress, and mastery validation.
Mint Green	`#34d399`	`text-emerald-300` / `bg-emerald-400`	Active lesson milestones, correct answers, neural sync indicators, and high mastery.
Dark Walnut Wood	`#25150c`	`border-[#25150c]`	Carved chalkboard exterior frame and wooden eraser base, anchoring university heritage.
Pine Slate Board	`#0e1713`	`bg-[#0e1713]`	Authentic dark green academic blackboard slate canvas for 120 FPS chalk derivations.
Warm Chalk Yellow	`#facc15`	`text-yellow-400`	Chalkboard lesson titles, Euler identities, active chalk tip cursor, and key theorem highlights.
Sky Blue Chalk	`#38bdf8`	`text-sky-400` / `border-cyan-400`	Geometric axes, cosine projections, secondary formulas, and Fourier wave curves.
Hologram Cyan	`#06b6d4`	`text-cyan-400` / `shadow-cyan-500/50`	3D Floating HUD borders, active index pointer rays, and brain wireframe pulses.
Academic Violet	`#8b5cf6`	`bg-violet-600` / `text-violet-400`	Speech waveform pill during active audio articulation and Research Suite tags.
Brass Gold	`#d97706`	`border-amber-600/60`	Blackboard corner brackets, executive commemorative plaques, and study XP badges.
Rose Coral	`#f43f5e`	`text-rose-400` / `bg-rose-500/20`	Negative marking penalty alerts, error traces, and complex plane imaginary axis projections.

Typography Stack

1. UI & Prose
Geist Sans / Inter. Neutral, geometric, high-legibility sans-serif optimized for multi-column dashboards and dense data tables.

2. Chalkboard Handwriting
Kalam & Caveat. Authentic cursive handwriting rendered stroke-by-stroke with varying thickness and chalk pressure.

3. Code & Mathematical Logic
Geist Mono & KaTeX. Monospaced code editor font and LaTeX math engine for quantum equations and algebraic derivations.

3. Core Cognitive Algorithms & Mathematical Formulations

BroadMind AI replaces arbitrary heuristics with established cognitive science and Bayesian probabilistic models:

A. Bayesian Knowledge Tracing (BKT) — Corbett & Anderson (1995)

Tracks the latent probability $P(L_t)$ that a student knows concept k after interaction step t:

> **Model Parameter Set:**
> • $P(L_0) = 0.25$: Prior probability of knowing concept before first attempt.
> • $P(T) = 0.15$: Transition / learning rate probability per opportunity.
> • $P(G) = 0.20$: Guess rate (probability of correct answer despite not knowing).
> • $P(S) = 0.10$: Slip rate (probability of incorrect answer despite knowing).

1. Posterior Evidence Update (Bayes' Theorem):

```
If Correct (obs = 1): P(L_{t-1} | obs=1) = [ P(L_{t-1}) * (1 - S) ] / [ P(L_{t-1}) * (1 - S) + (1 - P(L_{t-1})) * G ] If Incorrect (obs = 0): P(L_{t-1} | obs=0) = [ P(L_{t-1}) * S ] / [ P(L_{t-1}) * S + (1 - P(L_{t-1})) * (1 - G) ]
```

2. Learning Transition Step:

```
P(L_t) = P(L_{t-1} | obs) + [ 1 - P(L_{t-1} | obs) ] * P(T)
```

B. Ebbinghaus Memory Retention Decay Model

Human memory decays exponentially over elapsed time Δt (in hours) without active spaced rehearsal:

```
Decay Constant: lambda = ln(2) / (half_life_days * 24.0) Retained Mastery: P(L_retained) = P(L_t) * exp(-lambda * delta_t) Forgetting Risk: forget_risk = 1.0 - exp(-lambda * delta_t)
```

Standard Half-Life: $t_{ ext{half-life}} = 7.0 ext{ days}$ (dynamically fitted per learner as review history accumulates).

C. 5-Dimensional Cognitive Style Vector (Softmax Normalization)

Maintains a normalized probability vector representing the student's cognitive learning preference:

```
Style Vector: S = [ w_analogy, w_visual, w_stepByStep, w_formal, w_story ], where sum(w_i) = 1.0
```

- **Analogy**: Boosted when student clicks "explain differently" or rates metaphorical steps highly.
- **Visual**: Boosted by interactions with 3D anatomy, AC Oscilloscopes, and circuit diagrams.
- **Step-by-Step**: Boosted by progressive code hints and sequential algebraic proofs.
- **Formal**: Boosted by mathematical theorem queries and competitive PYQs.
- **Story**: Boosted by conversational audio replays and historical STEM vignettes.

D. Dynamic Complexity & Semantic Routing Algorithm

Incoming queries are parsed through an algorithmic heuristic to determine optimal model tier without waste:

TIER	TARGET MODEL	TRIGGER CONDITIONS	RESOURCE PROFILE
`flash_lite`	`nvidia/nemotron-3.5-bolt-e4b`	Single-fact recall, word count < 15, no math or code.	Max tokens: 1024, Temp: 0.7
`flash`	`nvidia/nemotron-3.5-bolt-e4b`	Standard academic doubts, 15–60 words, or student image upload.	Max tokens: 2048, Temp: 0.7
`pro`	`nvidia/nemotron-3.5-bolt-e4b` (Thinking)	Proofs, derivations, quantum/asymptotic terms, word count > 60.	Max tokens: 4096, Temp: 0.5

4. API Architecture & Multi-Provider Cascade Protocol

BroadMind AI implements a zero-downtime cascading fallback architecture ensuring no student ever encounters a frozen screen or brick-wall error:

> **Cascading Fallback Protocol**
> Primary: **NVIDIA NIM Nemotron 3.5 Bolt** (Sub-2.8s reasoning with thinking trace) →
> Fallback 1: **Groq Cloud** (`openai/gpt-oss-120b` , 500+ tok/s high-throughput) →
> Fallback 2: **OpenAI** (`gpt-4o-mini`) →
> Fallback 3: **BroadMind Deterministic Academic Engine** (Offline structured derivations).

How APIs are Formatted & Consumed

1. NVIDIA NIM (Reasoning & Code)
Endpoint: `https://integrate.api.nvidia.com/v1/chat/completions`
Model: `nvidia/nemotron-3.5-bolt-e4b`
Payload includes: `chat_template_kwargs: {"enable_thinking": True}` , `temperature: 0.5` , `seed: 42` . Returns reasoning thoughts and structured LaTeX math.

2. NVIDIA NIM (Vision & OCR)
Endpoint: `https://integrate.api.nvidia.com/v1/chat/completions`
Model: `minimaxai/minimax-m3.1-flash`
Payload accepts base64 data URIs: `{"type": "image_url", "image_url": {"url": "..."}}` . Solves diagrams and circuit schematics.

3. Sarvam AI (Indian Accent TTS)
Endpoint: `https://api.sarvam.ai/text-to-speech`
Model: `bulbul:v3`
Speakers: `aditya` (male), `priya` (female). Converts text to 22,050Hz WAV audio across English (Indian accent), Hindi, Tamil, Telugu, and 18 regional dialects.

4. DB_BM FastAPI Backend (Port 8001)
Endpoint: `http://localhost:8001/engine/*`
Endpoints: `POST /engine/log-event` , `GET /engine/state` , `GET /engine/next-step` , `POST /engine/route-query` , `DELETE /engine/user-data/{id}` . Dual PostgreSQL/SQLite.

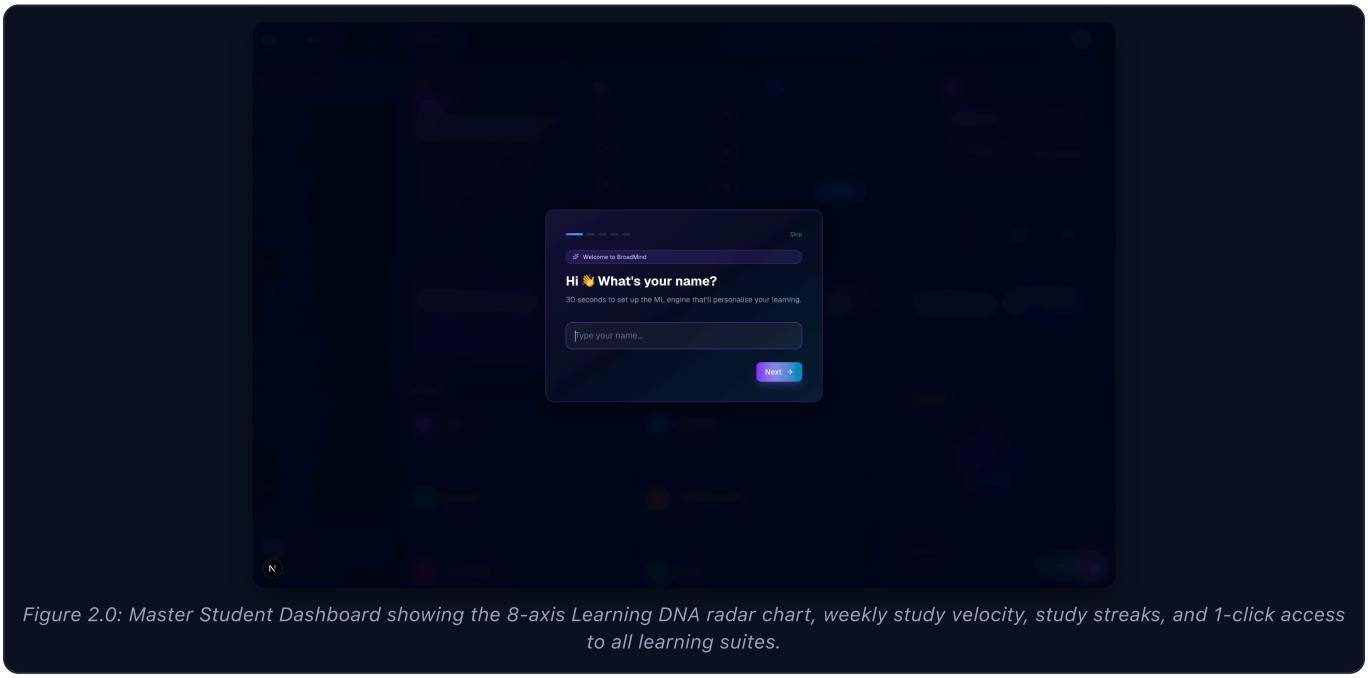


Figure 2.0: Master Student Dashboard showing the 8-axis Learning DNA radar chart, weekly study velocity, study streaks, and 1-click access to all learning suites.

5. Step-by-Step Module Engineering & Features Catalog

Below is the complete engineering specification for every module in BroadMind AI, outlining their step-by-step user process, internal methods, and options:

1. Avatar Tutor (Universal AI Professor)

Core Mission: World-class university lecturing on a 120 FPS high-DPI blackboard with living anatomical professor kinematics.

Step-by-Step Process: 1. Student selects curriculum masterclass (Euler's Identity, Newton's Gravity, Neural Networks) or types custom question.
2. NVIDIA NIM Nemotron plans a 3-step `AvatarTeachingPlan` with speech, hand gestures, emotions, and blackboard math.
3. Sarvam AI synthesizes Indian-accent voice audio; lips articulate in real-time phoneme visemes (`open_A` , `round_O` , `wide_E` , `viseme_M` , `teeth_TH`).
4. Blackboard jots character-by-character with chalk dust physics particles and diagram plots (unit circles, sigmoid curves).
5. Professor paces across stage (`walkX`), breathes (`bodyBob`), blinks every 3.5s, points to the 3D floating HUD, or contemplates proofs.

Options & Controls: Engine Mode (3D GLB vs 2.5D Living Canvas), View Mode (Studio, Board, Professor Focus), Action Deck (Pace, Write, Hologram, Ponder, Lecture), Emotion Deck (Eureka, Thinking, Lecture, Encouraging, Listening), Raise Hand Interruption button, Wooden felt eraser sweep wipe, 22 regional Indian languages.

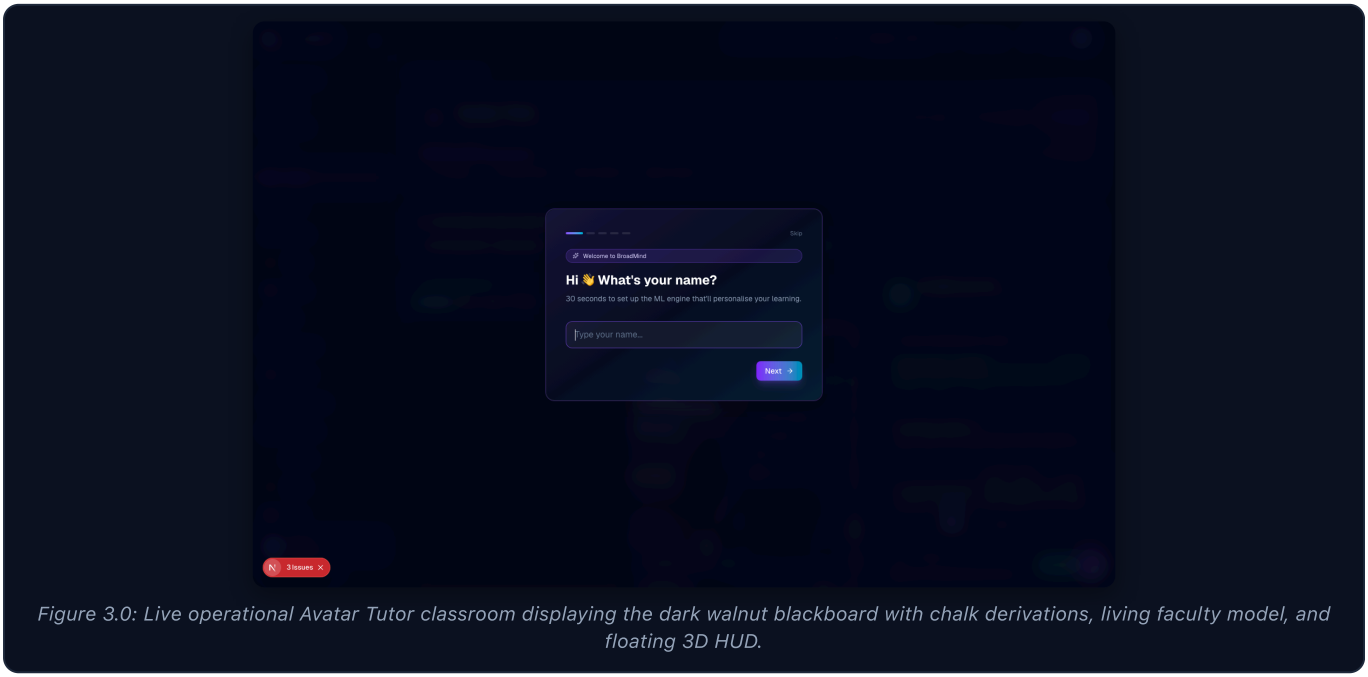


Figure 3.0: Live operational Avatar Tutor classroom displaying the dark walnut blackboard with chalk derivations, living faculty model, and floating 3D HUD.

2. Doubt Solver (Multimodal Step-by-Step)

Core Mission: Resolves complex STEM doubts with rigorous step-by-step LaTeX derivations and intuitive everyday Indian analogies.

Step-by-Step Process: 1. Student types query, uploads photo of textbook question, or draws on canvas.
2. If image present, Minimax 3.1 Flash OCR extracts equations; if text, Complexity Router selects model tier.
3. Returns structured output: (a) Core Principle, (b) Step-by-Step Proof, (c) Everyday Analogy (e.g. Cricket ball trajectory for parabola), (d) Verified Calculation Check.
4. Logs interaction to `DB_BM /interactions/` and updates concept BKT mastery.

Options: Text/Photo/Canvas input tabs, Model tier selector (NVIDIA / Groq / OpenAI), Difficulty pace, Language toggle, Copy LaTeX button.

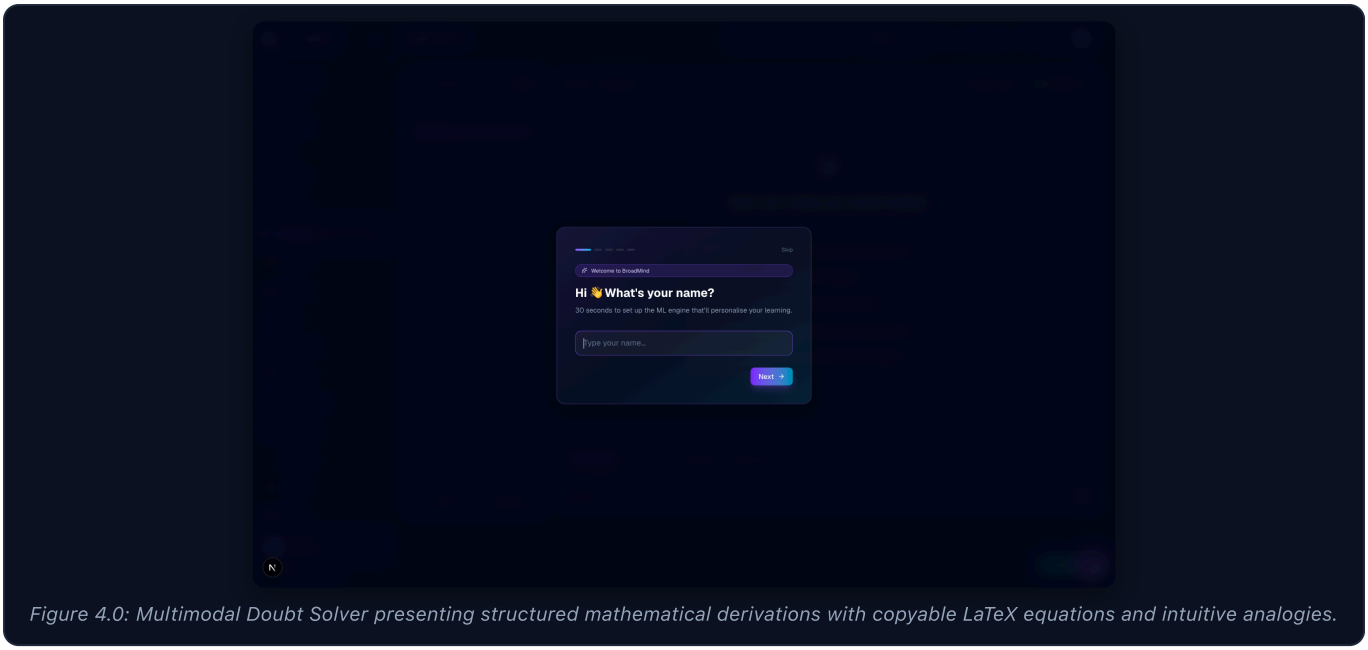


Figure 4.0: Multimodal Doubt Solver presenting structured mathematical derivations with copyable LaTeX equations and intuitive analogies.

3. Code Builder (Algorithm IDE & Test Harness)

Core Mission: Real-time algorithm coding suite with syntax-highlighted IDE, instant execution, and progressive hint ladders.

Step-by-Step Process: 1. Student selects coding problem (Dynamic Programming, Graph Traversal, Binary Search) or writes custom script.
2. Executes code locally in 26ms sandbox; runs assertions and test cases.
3. If syntax or logic error occurs, student requests progressive hints: Level 1 (Conceptual Clue), Level 2 (Pseudocode), Level 3 (Optimal Solution).
4. Logs run time, bug types, and time-to-fix to `DB_BM /learning-events/` to calibrate student problem-solving speed.

Options: Languages (Python, JavaScript, C++, Java), Hint depth level, Test case editor, Time complexity visualizer ($O(N)$ vs $O(N^2)$), Dark/Light syntax themes.

4. Recall Arena (Spaced Repetition & Dynamic MCQs)

Core Mission: Flashcards and timed quizzes powered by the Ebbinghaus forgetting model to solidify long-term memory retention.

Step-by-Step Process: 1. Student selects topic; engine queries `DB_BM /engine/state` to identify concepts with forgetting risk ≥ 0.30.
2. Generates dynamic 4-option MCQs and dual-sided flashcards.
3. Student flips card, rates recall confidence (1 to 5), or selects multiple-choice answer.
4. Engine updates BKT mastery and recalculates optimal next review date based on half-life decay.

Options: Study Mode (Flashcards vs Timed Quiz), Difficulty (Recall / Application / Reasoning), Sound effects toggle, Shuffle deck, Deck import/export.

5. Engineering Suite (Multidisciplinary Lab)

Core Mission: Interactive hardware and systems simulators for Electrical, Computer, Mechanical, and Civil Engineering.

Step-by-Step Process: 1. Student selects discipline tab (ECE, CSE, Mech, Civil).
2. In ECE: Manipulates Dual-Channel AC Oscilloscope knobs (Time/Div, Volts/Div, Frequency, Phase Offset) to observe real-time sinusoidal Lissajous figures.
3. In Logic Gates: Toggles digital switches (AND, OR, NAND, XOR) to trace truth table states.
4. In Civil/Mech: Tests beam load deflection formulas under varying structural forces.

Options: Oscilloscope Channel 1/Channel 2 selectors, Frequency dials (10Hz to 10kHz), Logic Gate truth table overlay, Beam cross-section selector.

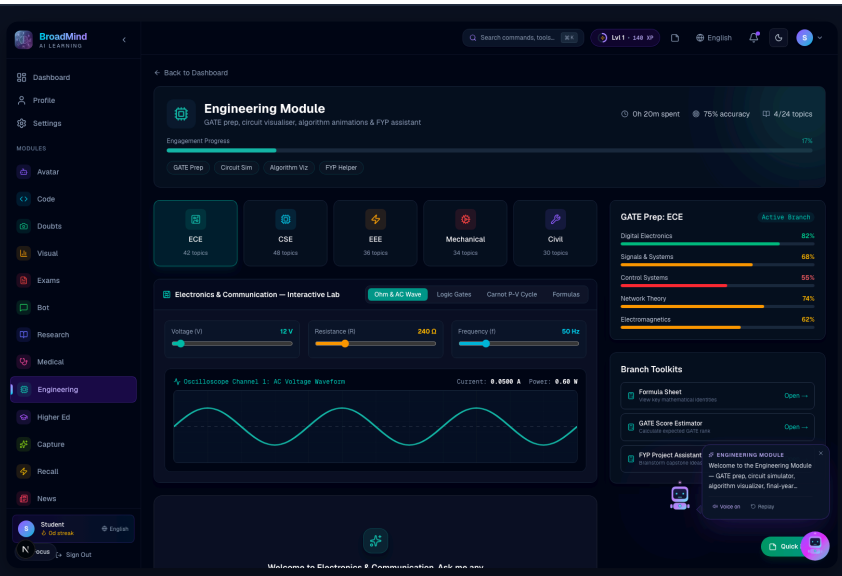


Figure 5.0: Engineering Suite showing the interactive dual-channel AC Oscilloscope with real-time waveform synthesis and Lissajous plotting.

6. Medical Education Suite (Clinical & Anatomy)

Core Mission: Clinical decision vignettes, pharmacological monographs, and interactive 3D anatomical organ models.

Step-by-Step Process: 1. Student selects organ system (Cardiovascular Heart, Neuroanatomy Brain, Ophthalmic Eye).
2. Explores anatomical structures with labeled pins and cross-sectional diagnostic views.
3. Solves clinical patient vignettes: analyzes symptoms, reviews vitals, orders differential lab tests, and selects drug dosage.
4. Reviews pharmacological monographs (mechanism of action, contraindications, adverse drug reactions).

Options: 3D anatomical rotation, Organ layer dissection (Muscular, Vascular, Neural), Clinical pearl reveal, Monograph PDF export.

7. Exam Assistant & Competitive Indian Exams Directory

Core Mission: Timed mock exam cockpit for GATE, JEE Advanced, UPSC Civil Services, and NEET with negative marking analytics.

Step-by-Step Process: 1. Student selects target exam and mock test paper.
2. Starts timed exam with official countdown clock, question palette, and mark-for-review tags.
3. Submits paper; engine calculates score, accuracy, negative marks deducted ($- rac{1}{3}$ for GATE), and topic-wise mastery.
4. Exams Directory tracks official exam notification dates, eligibility criteria, and syllabus PDF links.

Options: Negative marking calculator toggle, Category filter (Engineering, Medical, Civil Services, Law), Starred bookmarks, PYQ year filter (2018–2025).

8. Study Planner (Intelligent Spaced Scheduler)

Core Mission: Generates adaptive daily and weekly study timetables prioritizing weak concepts and upcoming forgetting risks.

Step-by-Step Process: 1. Student enters target exam date and daily study availability (e.g. 4 hours/day).
2. Engine queries `DB_BM /engine/next-step` to extract concepts needing review or practice.
3. Allocates balanced study blocks with integrated Pomodoro breaks (25 min focus, 5 min rest).
4. Student checks off completed tasks; missed blocks automatically roll over without breaking streak count.

Options: Daily/Weekly calendar view, Pomodoro timer with ambient chime, Target exam countdown widget, One-click reschedule.

9. Research Suite & Academic Paper Explorer

Core Mission: Curated academic paper explorer interfacing arXiv and OpenAlex for advanced literature reviews.

Step-by-Step Process: 1. Student searches topic (e.g. "Quantum Graph Neural Networks in VLSI").
2. Fetches matching peer-reviewed papers with abstracts and open-access PDF links.
3. AI extracts Key Findings, Methodology, and Limitations in bulleted executive briefs.
4. Generates formatted academic citations in APA 7th edition, IEEE transactions, and BibTeX.

Options: Citation style picker (APA, IEEE, BibTeX), Date range filter, Direct arXiv PDF viewer, Saved papers portfolio.

10. Capture & Notes Hub

Core Mission: Multimodal lecture note-taking integrating voice dictation, OCR image extraction, and markdown summarization.

Features: Real-time speech-to-text dictation, camera photo OCR scanning, auto-generated executive study summaries, tagging by concept taxonomy.

11. Daily Student News & Current Affairs

Core Mission: High-yield current affairs curated specifically for Indian competitive exams (UPSC GS-I/II/III/IV, RBI Grade B).

Features: Syllabus relevance tags (Polity, Economy, S&T, Ecology), 60-word quick editorial briefs, daily quiz questions based on today's headlines.

12. Omni-Bot (Multi-Channel Academic Companion)

Core Mission: Omnipresent conversational bot maintaining student personality and state across Web, WhatsApp, and Voice.

Features: Multi-channel dispatch routing, persistent memory across pages, instant formula lookups, quick doubt routing.

13. STEM Visualisation Lab

Core Mission: Interactive visual sandbox for abstract mathematical and computer science phenomena.

Features: Dynamic Fourier square-wave harmonic synthesis, Binary Search Tree (BST) visual insertions and rotations, RC filter frequency response curves.

14. Learning Insights (Learning DNA Radar)

Core Mission: Comprehensive cognitive analytics dashboard visualizing the student's holistic learning style.

Features: 8-axis Learning DNA radar (Mastery, Retention, Velocity, Focus, Problem-Solving, Visual, Step-by-Step, Formal), daily study minute heatmaps.

15. Higher Education & UGC Syllabus Explorer

Core Mission: University semester unit breakdown based on official UGC and AICTE model curricula.

Features: Semester-wise subject trees, unit-by-unit learning outcomes, recommended standard textbooks, prerequisite dependency maps.

16. User Profile, Authentication & Settings

Core Mission: Student onboarding, credential management, localization, and privacy preferences.

Features: Email/Password, Phone number, and Guest login; HTTP-only secure cookie session; 22 official Indian regional languages; Dark/Light theme toggle.

6. Data Governance, Security & DPDP Act 2023 Compliance

> **Indian Digital Personal Data Protection (DPDP) Act 2023 Conformance**
> - **Pseudonymous Telemetry**: All events in `learning_events` are keyed to random UUIDs with no personal identifiable info (PII).
> - **Right to be Forgotten**: Instant complete data purge via `DELETE /engine/user-data/{user_id}`.
> - **Parental Consent**: Verification gates for minor students under age 18 before persistent telemetry collection.